G.C.E. (O/L) · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 01

Mathematics I

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Time: Two hours **Total marks:** 100

Index number: .....

Important:

- Answer **all** questions on this question paper itself.
- Write your answers, and the working that leads to them, only in the space provided under each question.
- Show the relevant steps and write the correct units when answering.
- Marks: Part A – 2 marks for each question; Part B – 10 marks for each question.
- Scrap paper may be obtained for rough calculations.

Part	A (1 – 25)	B1	B2	B3	B4	B5	Total
Marks	/50	/10	/10	/10	/10	/10	/100

Part A

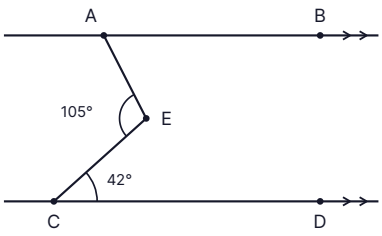
Answer all questions on this paper itself. (2 marks each) Take π = 22/7.

1. Kasun's farm has a field that 12 workers, all working at the same rate, can weed in 15 days. How many more workers are needed to weed the field in 9 days?

Answer: .....

2. In the figure, AB ∥ CD. Find the size of BÂE.

Answer: .....



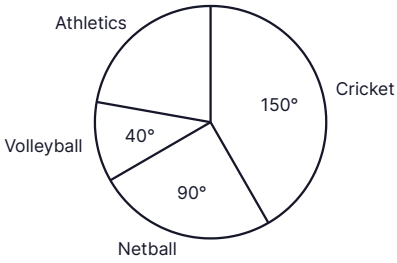
3. Simplify: $\frac{3}{4x} + \frac{5}{6x} - \frac{1}{3x}$

Answer: .....

4. A right prism has a triangular base. Which one of the following quadrilaterals can **not** be a face of this prism? Put a tick (✓) against it.
(i) a square (ii) a rhombus which is not a square (iii) a rectangle

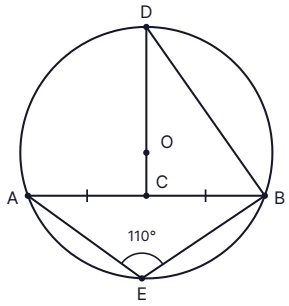
Answer: .....

5. The pie chart shows the favourite sport of the students in a class. Each student chose exactly one sport. 12 students chose volleyball. How many students chose cricket?



Answer: .....

6. In the figure, O is the centre of the circle. The chord AB is bisected at C and DOC is a straight line. If $A\hat{E}B = 110°$, find $C\hat{D}B$.



Answer: .....

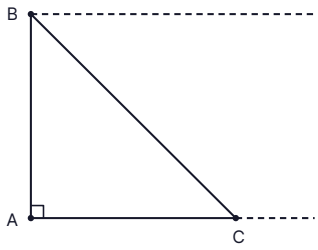
7. Factorise: $3x^2 - 75$

Answer: .....

8. It is given that $10^{0.6990} = 5$. Write the value of lg 50.

Answer: .....

9. AB is a vertical lamp post standing on level ground (BÂC = 90°). Kasun at C on the ground looks up at the lamp B, and a worker at B looks down at Kasun. On the figure, mark the angle of elevation of B from C as a and the angle of depression of C from B as b, and write the relation between a and b.



Answer: .....

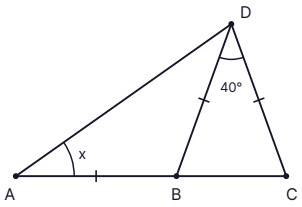
10. The height of a cylindrical can is 4 times its base radius r. The curved surface area of a cylinder of base radius r and height h is $2\pi rh$. Find the ratio of the area of paper needed to cover the curved surface of the can to the area of paper needed to cover its circular base.

Answer: .....

11. Two fair tetrahedral dice have their four faces marked 2, 4, 6 and 8. Both dice are rolled once and the number on the face on which each die rests (the bottom face) is noted. Find the probability that the sum of the two numbers is 12.

Answer: .....

12. In the figure ABC is a straight line and AB = BD = DC. If BD̂C = 40°, find the value of x.



Answer: .....

13. The universal set is ξ = {1, 2, 3, ..., 10}. A = {factors of 12}, B = {even numbers}. Tick (✓) the correct statement.
(i) A ∩ B = {2, 4} (ii) A ∪ B = {1, 2, 3, 4, 6, 8, 10} (iii) n(A′) = 4

Answer: .....

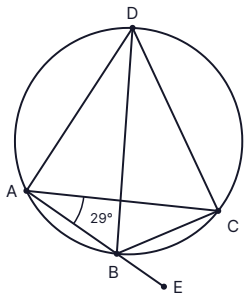
14. Simplify: $\frac{6x^3y^2 \times 5xy^3}{10x^2y^2}$

Answer: .....

15. The second term and the third term of a geometric progression are −12 and 6 respectively. Find the fifth term.

Answer: .....

16. In the figure, ABCD is a cyclic quadrilateral. BD bisects $A\hat{D}C$ and AB is produced to E. If $B\hat{A}C = 29°$, find $C\hat{B}E$.



Answer: .....

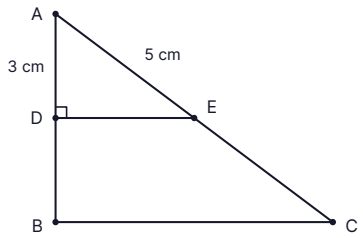
17. Find the highest common factor of $24p^3q^2$, $36p^2q^3$ and $60p^2q$.

Answer: .....

18. A right prism whose base is a triangle of area 154 cm^2 and a right circular cylinder have the same height and the same volume. Find the base radius r of the cylinder. (Volume of a cylinder = $\pi r^2 h$)

Answer: .....

19. In the triangle ABC, D and E are the midpoints of AB and AC respectively, and $A\hat{D}E = 90°$. AD = 3 cm and AE = 5 cm. Find the perimeter of the trapezium BCED.

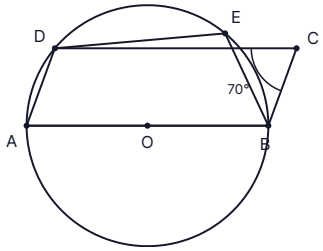


Answer: .....

20. Find the equation of the straight line passing through the points (2, 3) and (6, 5).

Answer: .....

21. In the figure, O is the centre of the circle, AOB is a straight line and ABCD is a parallelogram. If BĈD = 70°, find DÊB.

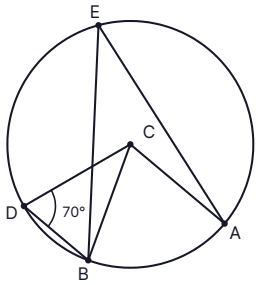


Answer: .....

22. Nadeesha invested a sum of money for 2 years at an annual compound interest rate of 5%. After 2 years the total amount she received was Rs 22 050. Find the sum she invested.

Answer: .....

23. In the figure, C is the centre of the circle and AC ∥ BD. If BD̂C = 70°, find AÊB.

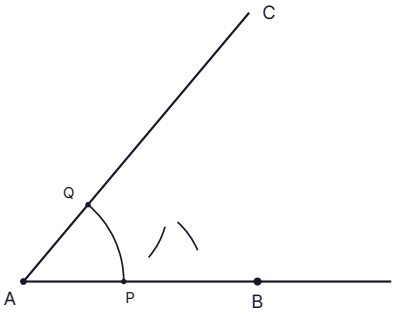


Answer: .....

24. Solve the inequality $4x + 15 > 3$ and write the negative integer values of x that satisfy it.

Answer: .....

25. Two paths AB and AC meet at A. A circular pond touches the path AB at B and also touches the path AC. The figure shows part of the construction for finding the centre of the pond. Using a straight edge and a pair of compasses, complete the construction and mark the centre O of the pond.



Answer: .....

Part B

Answer all questions on this paper itself. (10 marks each) Take π = 22/7 where needed.

1. Dilini, a teacher, pays income tax at 10% on the part of her monthly salary that exceeds Rs 75 000. After paying the tax, she spends $\frac{1}{5}$ of the money she receives on food, and $\frac{5}{8}$ of the remainder on other expenses.

(i) What fraction of the money she receives is left after buying food? (Marks 1)

(ii) What fraction of the money she receives is left after spending on both food and other expenses? (Marks 2)

(iii) The money left after both expenses is Rs 36 000. Find the amount she spends on food and the amount she spends on other expenses. (Marks 2)

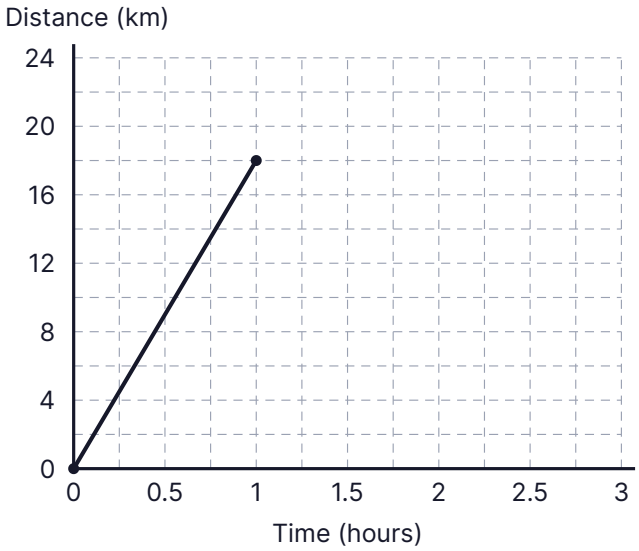
(iv) Find her monthly salary before tax. (Marks 3)

(v) Suppose the tax-free limit is raised so that she pays no income tax at all, while the amount she spends on food stays the same. What percentage of her salary is spent on food? (Marks 2)

2. **(a)** Ravi cycles from his home to Kasun's farm at a uniform speed. The distance–time graph for the first hour of his journey is drawn on the grid below. He stays at the farm for 30 minutes and then cycles back home along the same route at a uniform speed, taking 1½ hours.

(i) Complete the distance–time graph for the whole journey. (Marks 3)

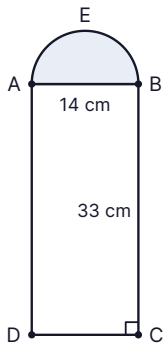
(ii) Find his speed on the return journey in km h^{-1}. (Marks 2)



(b) The figure (not drawn to scale) shows a plane figure made of a semicircle AEB with diameter AB = 14 cm and a rectangle ABCD with BC = 33 cm and $B\hat{C}D = 90°$. Take $\pi = \frac{22}{7}$.

(i) Find the perimeter of the figure. (Marks 2)

(ii) The area of the rectangle is how many times the area of the semicircle? (Marks 3)



3. Kasun rents out a shop for Rs 12 500 per month and receives the rent for the whole year in a single payment. The Municipal Council has assessed the annual value of the shop at Rs 50 000 and charges annual rates of 10% of the assessed value. Kasun spends 10% of the year's rent on maintenance. He then adds some more money to what is left after paying the rates and the maintenance, and buys shares of Rs 80 each. Each share pays an annual dividend of 7.5% of its value, and he receives Rs 15 000 as dividends for the year.

(i) Find the annual rates he has to pay. (Marks 2)

(ii) Find the amount of money left after paying the rates and the maintenance. (Marks 2)

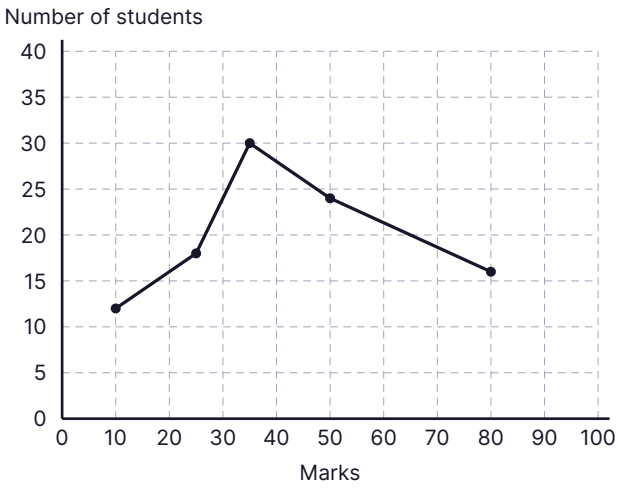
(iii) Find

(a) the number of shares he bought, (Marks 3)

(b) the extra money he added, as a percentage of the total amount he invested in shares. (Marks 3)

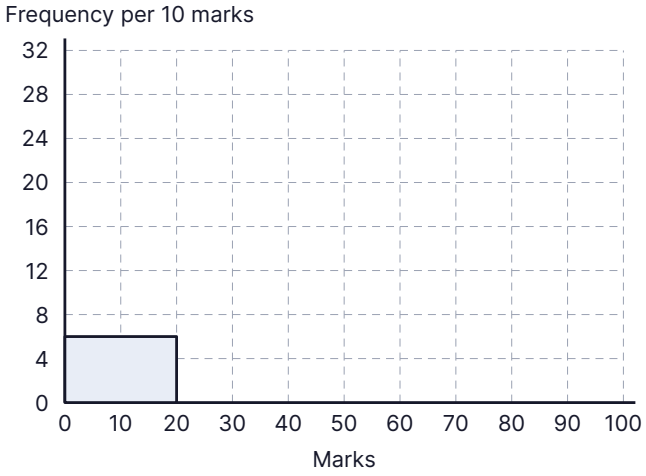
4. The frequency polygon shows the marks obtained in a Mathematics test by 100 students of Kasun's school. The points are plotted at the mid-values of the classes. An incomplete frequency table is also given. A class such as 20 – 30 includes 20 but not 30.

Marks	Number of students	Cumulative frequency
0 – 20	12	12
20 – 30	.....	.....
30 – 40	.....	.....
40 – 60	.....	.....
60 – 100	.....	.....



(i) Using the frequency polygon, complete the "number of students" column of the table. (Marks 3)

(ii) The classes are not of equal width, so in a histogram the height of each bar is the frequency per 10 marks (frequency ÷ class width × 10). The bar for the class 0 – 20 is already drawn. Complete the histogram on the grid below. (Marks 3)



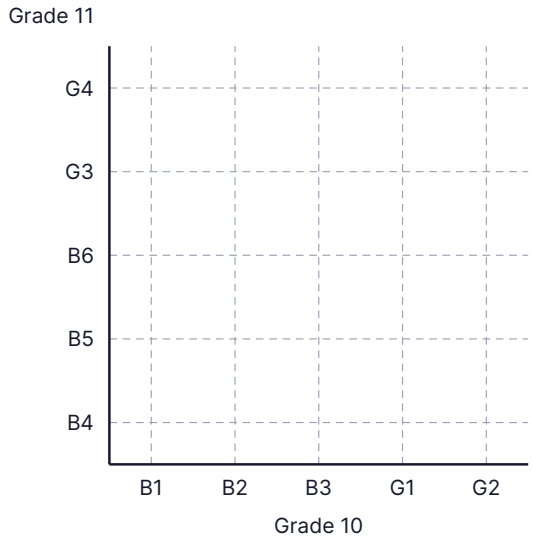
(iii) Complete the cumulative frequency column of the table. (Marks 2)

(iv) One student is chosen at random. Find the probability that the student scored 40 marks or more. (Marks 2)

5. For a relay team, Nadeesha, the sports teacher, chooses one student at random from the Grade 10 squad and one student at random from the Grade 11 squad. The Grade 10 squad has three boys (B1, B2, B3) and two girls (G1, G2). The Grade 11 squad has three boys (B4, B5, B6) and two girls (G3, G4). The grid shows the squads; the sample space is to be marked on it.

(i) Mark the sample space of this selection on the grid with the symbol '×'. (Marks 2)

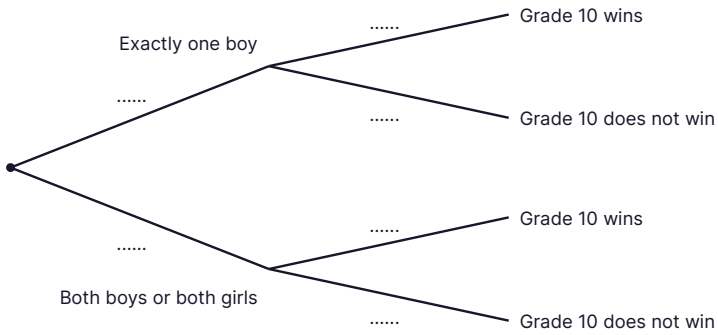
(ii) Enclose, with closed curves, the outcomes in which exactly one boy is selected, and find the probability of this event. (Marks 3)



If exactly one boy is selected, the probability that Grade 10 wins the relay is $\frac{3}{5}$. If two boys or two girls are selected, the probability that Grade 10 wins is $\frac{2}{5}$.

(iii) Complete the probabilities on the tree diagram. (Marks 3)

(iv) Find the probability that Grade 10 does not win. (Marks 2)



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